

THE GEMINI PROTOCOLS — CHAPTER 302

PHASE 302: THE TAP DRONE

the machine that flies where there is no air — on the gas you were going to throw away

Nothing hovers on the moon. No rotor bites, no wing lifts, no balloon floats. Anything that leaves the ground there leaves it on reaction — so we built the flyer the vacuum allows.

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STANDARD NOTATION — MATERIALS & CALIBRATIONS: Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice. Our designs are base calibrations to existing strength materials: no speed, load or pressure exceeds current basic materials capability. We do not spec methods, and we do not teach welders how to weld.

1 • The Law

On the moon there is no air. Nothing wings, nothing hovers on a rotor, nothing glides. Anything that leaves the ground leaves it on reaction. The tap drone is the flyer built from that law: a cold-gas rocket that jumps — and taps to stay up.

2 • The Gas Is Free

Rockets are fuel — but CO₂ is a byproduct. The crew breathes it out, the scrubbers pull it out, and life support pays to be rid of it. A cold-gas CO₂ drone drinks waste (cold-gas CO₂, ~640 m/s exhaust class). You only have so much oxygen, so you never spend it searching: the drones fly on the gas you were going to throw away.

3 • Two Classes

The tap drone is built in two classes. The long-distance class: range machines, ballistic hoppers that scout and relay far ahead of the buggy, riding the Phase 211 corridor maps. The poppers: small, carried in numbers, and mandatory — the backup that is never left behind.

4 • The Popper — Mandatory Backup

When you lose radio, or have no signal, you pop these up for vision and signal. First a quick burst up; then the drone taps to keep itself up — taps like tapping a balloon. No gas spent hovering: it auto-taps whilst you look around with the camera. It carries strobe lights for when you are a long way out. A buggy that rides out without poppers is one radio fault away from blind — so the poppers ride or the buggy does not.

5 • The Physics of the Tap

The audit's note, seated with the doctrine: tapping does not beat gravity. While the drone is airborne, the average tap impulse equals the drone's weight — 1.62 m/s^2 lunar, every second it is up. Pulsing buys simplicity, instant throttle, and free gas — not gravity. Sized honestly, the tap is cheap and the bottle lasts; sized on wishful thinking, it is a fountain that dies. The numbers go to the bench, not the brochure.

6 • Operations

Standard popper work: lost-signal search and relay — burst up, auto-tap, camera live, strobe on if the buggy is far. Long-distance class scouts the corridor ahead and relays over the horizon. Recovery support: a popped drone marks the stricken buggy's position and strobes it for the winch-rocket line. Every gas fill is logged against the scrubber output — the drones fly on waste, and waste is still mass.

7 • Materials

Pressure-class CO_2 bottles, cold-soak rated valves, graphite grease per the lubricant law (Tru-Blu nevermelt bentonite class, matched to medium) on every moving mechanism. Structure light, nozzle plain, camera and strobe hardened for dust and the two-week night.

Owed to the Bench

Tap-impulse equivalence test — measured average impulse against airborne weight. Popper tap-rate and bottle sizing — taps per flight, gas per tap, flights per fill. Valve cold-soak cycling across the two-week night. These are owed numbers, and until the bench pays them the tap drone flies on estimates.

THE SCALP: everything else on the moon is grounded — the tap drone is the only thing that flies, and it flies on waste.